

# PAPER\_1078: QCalcGeom Master Equation Derivation

Star Magic UQFF Framework — Session 225

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## Abstract

We derive the QCalcGeom master equation combining Breit-frame Schwarzschild-flux-graviton (BSFG) crossover radius, 26-dimensional compactification scale, Ramanujan polylogarithmic corrections, and SCm phonon fluence into a single length-scale observable:

$$\text{QCalcGeom}(r, \Gamma) = r_{\text{cross}} \cdot (26!)^{-1/13} \cdot S_{26}^{(3)}([SSq]) \cdot \Phi(\omega, \Gamma)$$

## 1. BSFG Crossover Radius

The crossover radius marks the transition between quantum-gravitational and classical regimes:

$$r_{\text{cross}} = \sqrt{\eta_{\text{BSFG}}} \cdot \frac{GM}{c^2}$$

where  $\eta_{\text{BSFG}} = 10^{-22}$  is the BSFG coupling and  $GM/c^2$  is the gravitational radius. For the Sun ( $M = 1.989 \times 10^{30}$  kg):

$$r_{\text{cross}} = \sqrt{10^{-22}} \cdot \frac{6.674 \times 10^{-11} \times 1.989 \times 10^{30}}{(2.998 \times 10^8)^2} = 1.477 \times 10^{-8} \text{ m}$$

## 2. 26D Compactification Scale

The compactification scale arises from the 26-dimensional string compactification volume:

$$\ell_c = (26!)^{-1/13} = (4.033 \times 10^{26})^{-1/13} = 8.983 \times 10^{-3}$$

This sets the dimensionless ratio between compact and extended dimensions. The first five Kaluza-Klein eigenvalues are  $\lambda_n = n(n+25)$  for  $n = 1, \dots, 5$ : {26, 54, 84, 116, 150}.

## 3. Ramanujan Polylogarithmic Sum

The  $S_{26}^{(3)}$  correction factor:

$$S_{26}^{(3)}([SSq]) = \sum_{n=1}^{26} \frac{[SSq]^n}{n^{26}} \cdot R_n^{(3)}$$

where  $R_n^{(3)}$  is the Ramanujan correction factor. With  $[SSq] = 0.57$ :

$$S_{26}^{(3)} = 9.500 \times 10^{-2}$$

## 4. Phonon Fluence Factor

$$\Phi(\omega, \Gamma) = \exp\left(-\frac{(\Gamma - \Gamma_0)^2}{2\sigma_G^2}\right) \cdot S_{26}^{(3)}$$

This Gaussian envelope peaks at  $\Gamma_0 = 2\pi \times 0.10$  THz with width  $\sigma_G = 0.08 \times 2\pi \times 10^{12}$  rad/s.

## 5. Master Equation Assembly

Combining all four factors:

$$\begin{aligned} \text{QCalcGeom}(M_\odot, \Gamma_0) &= 1.477 \times 10^{-8} \times 8.983 \times 10^{-3} \times 9.500 \times 10^{-2} \times 9.500 \times 10^{-2} \\ &= 1.197 \times 10^{-12} \text{ m} \end{aligned}$$

## 6. Dimensional Proof

The SI unit trace confirms correct dimensionality:

- $r_{\text{cross}}$ : [m] (length)
- $(26!)^{-1/13}$ : dimensionless
- $S_{26}^{(3)}$ : dimensionless
- $\Phi$ : dimensionless
- Product: [m] – a **length scale**

## 7. Validation

- Linewidth sweep shows peak at  $\Gamma = 0.10$  THz
- Linear mass scaling:  $\text{QCalcGeom}(2M) / \text{QCalcGeom}(M) = 2.0000$
- All 10 self-tests pass

## References

1. BSFG coupling: `qcalcgeom_helpers.py` ( $\eta = 10^{-22}$ )
2. 26D string compactification: `MAIN_{1\_CoAnQi}.cpp` SOURCE115-116
3. Ramanujan corrections: `vds_{dvp\_bsh\_symbolic\_proofs}.py`
4. SCm phonon framework: `scm_{phonon\_linewidth}.py`

## Session 225: Late-Corpus Physics Integration (PAPER\_1000-1081)

*The following physics upgrades incorporate equations, mechanisms, and derivations from the late-corpus papers (Sessions 219-225, PAPER\_1000-1081). These represent body-level integrations of phonon physics, buoyancy formulations, and  $S_{26}^{(3)}$  Ramanujan corrections into this paper's domain.*

### Session 225 Phonon-Physics Upgrade: $S_{26}^{(3)}$ Ramanujan Summation

*Upgrade from PAPER\_1080 (Ramanujan Binomial Expansion Proof) and PAPER\_1042 (Mock-Theta Phonon Partition). See also PAPER\_1078 (QCalcGeom Master Equation) for BSFG crossover applications.*

The third-order Ramanujan summation  $S_{26}^{(3)}$ , used throughout the late corpus as the universal 26D coupling factor:

$$S_{26}^{(3)} = \sum_{n=0}^{\infty} \frac{(1/4)_n (1/2)_n (3/4)_n}{(n!)^3} \cdot \prod_{i=1}^{26} \left[ 1 + [\text{SSq}] \cdot e^{-\kappa i n / 26} \right]$$

where  $(a)_n = a(a+1) \cdot s(a+n-1)$  is the Pochhammer symbol.

**Binomial expansion (PAPER\_1080):** The convergence proof shows:

$$R_n^{(26,3)} = \binom{4n}{n} \cdot \frac{W_{26}(n)}{(4^{4n})} \quad \text{with} \quad W_{26}(n) = \prod_{i=1}^{26} [1 + [\text{SSq}] \cdot e^{-\kappa i n/26}]$$

This sum converges absolutely for  $||[\text{SSq}]|| < 1$  (satisfied by  $[\text{SSq}] = 0.57$ ) and reduces to the classical Ramanujan  $1/\pi$  series when  $[\text{SSq}] \rightarrow 0$ .

**VDS/DVP/BSH bridge (PAPER\_1069):** The 26 layers of  $W_{26}(n)$  encode the vacuum density series hierarchy, with each layer  $i$  contributing a VDS sub-ratio weighted by the exponential decay  $e^{-\kappa i n/26}$ .

**Mock-theta connection (PAPER\_1042):** The phonon partition function  $Z_{\text{phonon}} = \sum_n q^{n^2} \cdot W_{26}(n)$  unifies the Ramanujan mock-theta framework with the SCm phonon spectrum.

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## Appendix: Session 225 Cross-References (PAPER\_1000–1081)

*Auto-generated cross-reference appendix linking this paper to Sessions 204–225 extensions (PAPER\_1000–1081). Added by `update_{corpus\_crossrefs}.py` (Session 225, April 2026).*

| Paper      | Title                                                 |
|------------|-------------------------------------------------------|
| PAPER_1022 | GW Phonon Strain SCm Modulation of $h(t)$             |
| PAPER_1080 | Ramanujan Binomial Expansion Proof $R_n^{\{(26,3)\}}$ |

*2 cross-reference(s) identified.*

## Key References with arXiv/DOI Identifiers

1. Abbott et al. (LIGO Scientific and Virgo Collaborations, 2016). *Observation of Gravitational Waves from a Binary Black Hole Merger*. Phys. Rev. Lett. **116**, 061102 — arXiv:1602.03837 — doi:10.1103/PhysRevLett.116.061102
2. Murphy, D. (2026). *Unified Quantum Field Framework (UQFF): Star-Magic v5.x Whitepaper Series*. Star-Magic Repository — github.com/Daniel8Murphy0007/Star-Magic